

- m Compatible quality of service mechanisms: Quality of Service (QoS) for networking is essential when transferring high-definition media streams in the digital home, particularly in the presence of best effort traffic. For this to work, vendors must agree on how to address QoS in the digital home; but devices must still interoperate even if there are no QoS mechanisms implemented.
- m Compatible authentication and authorisation mechanisms for users and devices: A number of authentication and authorisation mechanisms are being considered by device manufacturers and application developers to provide appropriate security for access and control. It is imperative to settle on a compatible authentication and authorisation framework to enable devices to request and/or grant access to particular devices and services in the home.

Additional Elements

In addition to the building blocks described above, other capabilities and issues need to be investigated and addressed for the digital home. Following are important capabilities that the DHWG will investigate further.

Digital Rights Management/Content Protection

In order for premium digital content to be made available for use with digital home devices, content providers understandably insist that their content be protected from unauthorised copying and use. At the same time, consumers expect to be able to store, transport, and use that content at any location and on any device within their digital home. Balancing the providers' need for protection and the consumers' fair use rights and expectations, while providing interoperability between all networked devices that might handle the content, is a complex problem. Content protection methods must also be user friendly.

Today, there are several Digital Rights Management (DRM) technologies available to designers and content providers. One or more of these solutions will typically be provided on digital home devices to protect, administer, and distribute stored content, as one component of content protection in the digital home. Other components of DRM that support additional user scenarios are being considered for development in the UPnP Forum and elsewhere in the industry.